PLU

Descomponer una matriz en dos donde una es triangular inferior

$PA = LU$ donde A es la matriz

P es una matriz de permutaciones

U una matriz triangular superior

L " " " inferior

Ejemplo:

$$A = \begin{bmatrix} 0 & 0 & 2 \\ -1 & 5 & -2 \\ 3 & 6 & 7 \end{bmatrix} \qquad PA = LU$$

$$U_0 = \begin{bmatrix} 0 & 0 & 2 \\ -1 & 5 & -2 \\ 3 & 6 & 7 \end{bmatrix} \xrightarrow[F_3 \to F_1]{} \begin{bmatrix} 3 & 6 & 7 \\ -1 & 5 & -2 \\ 0 & 0 & 2 \end{bmatrix} \xrightarrow[F_2 + \frac{1}{3} F_1]{} \overbrace{\begin{bmatrix} 3 & 6 & 7 \\ 0 & 7 & 1/3 \\ 0 & 0 & 2 \end{bmatrix}}^{U}$$

permutación

triangular superior

$$L_0 = \begin{bmatrix} 0 & 0 & 0 \\ -1/3 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow L = \begin{bmatrix} 1 & 0 & 0 \\ -1/3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$P_0 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \xrightarrow[F_3 \to F_1]{} \underbrace{\begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}}_{P}$$

$$PA = LU \Rightarrow \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 2 \\ -1 & 5 & -2 \\ 3 & 6 & 7 \end{pmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -1/3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3 & 6 & 7 \\ 0 & 7 & 1/3 \\ 0 & 0 & 2 \end{bmatrix}$$

Ejemplo 2:

$$A = \begin{pmatrix} 2 & -2 & 1 \\ -8 & 11 & 5 \\ 4 & -13 & 2 \end{pmatrix}$$ Buscar la PLU

$$U_0 = \begin{bmatrix} 2 & -2 & 1 \\ -8 & 11 & 5 \\ 4 & -13 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & -2 & 1 \\ 0 & 3 & 9 \\ 0 & -9 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & -2 & 1 \\ 0 & -9 & 0 \\ 0 & 3 & 9 \end{bmatrix} \rightarrow \boxed{\begin{bmatrix} 2 & -2 & 1 \\ 0 & -9 & 0 \\ 0 & 0 & 9 \end{bmatrix} = U}$$

$F_3 - 2F_1$ $\qquad$ $F_3 \leftrightarrow F_2$ $\qquad$ $F_3 - (-\frac{1}{3})F_2$

$F_2 + 4F_1$ // $F_2 - (-4)F_1$

$$L_0 = \begin{bmatrix} 0 & 0 & 0 \\ -4 & 0 & 0 \\ 2 & 0 & 0 \end{bmatrix} \xrightarrow{F_3 \leftrightarrow F_2} \begin{bmatrix} 0 & 0 & 0 \\ 2 & 0 & 0 \\ -4 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 0 & 0 \\ 2 & 0 & 0 \\ -4 & -\frac{1}{3} & 0 \end{bmatrix} \Rightarrow \boxed{L \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -4 & -\frac{1}{3} & 1 \end{bmatrix}}$$

$$P_0 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow \boxed{P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}}$$

Aplicación:

$$Ax = b \Rightarrow x = A^{-1}b$$ → costoso computacionalmente

$$\begin{cases} 2x_1 + 3x_2 + 4x_3 = 6 \\ 4x_1 + 5x_2 + 10x_3 = 16 \\ 4x_1 + 8x_2 + 2x_3 = 2 \end{cases} \Rightarrow A = \begin{pmatrix} 2 & 3 & 4 \\ 4 & 5 & 10 \\ 4 & 8 & 2 \end{pmatrix} \quad b = \begin{pmatrix} 6 \\ 16 \\ 2 \end{pmatrix}$$

$$\det(A) = (20 + 16 \cdot 8 + 120) - (20 + 160 + 12)$$

$$= 128 + 120 - 160 - 12 \neq 0$$

∴ compatible determinado

Sea $PA = LU \rightarrow Ax = b$

$$PAx = Pb$$

$$LUx = Pb$$

Sea $z = Ux$ nuevo vector de incognitas

$$\tilde{b} = Pb$$

$$\underbrace{L}_{\text{Triangular inferior}} z = \tilde{b}$$

$$U_0 = \begin{pmatrix} 2 & 3 & 4 \\ 4 & 5 & 10 \\ 4 & 8 & 2 \end{pmatrix} \rightarrow \begin{pmatrix} 2 & 3 & 4 \\ 0 & -1 & 2 \\ 0 & 2 & -6 \end{pmatrix} \rightarrow \begin{pmatrix} 2 & 3 & 4 \\ 0 & -1 & 2 \\ 0 & 0 & -2 \end{pmatrix}$$

$F_3 - 2F_1$ $\quad$ $F_3 - (-2)F_2$

$F_2 - 2F_2$

$$L_0 = \begin{pmatrix} 0 & 0 & 0 \\ 2 & 0 & 0 \\ 2 & -2 & 0 \end{pmatrix} \Rightarrow L = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 2 & -2 & 1 \end{pmatrix} \qquad P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\underbrace{P}_{I}A = LU \Rightarrow A = LU$$

$$L\underbrace{Ux}_{z} = b \Rightarrow Lz = b \qquad z = \begin{pmatrix} z_1 \\ z_2 \\ z_3 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 2 & -2 & 1 \end{pmatrix} \begin{pmatrix} z_1 \\ z_2 \\ z_3 \end{pmatrix} = \begin{pmatrix} 6 \\ 16 \\ 2 \end{pmatrix} \Rightarrow \begin{cases} z_1 = 6 \\ 2z_1 + z_2 = 16 \\ 2z_1 - 2z_2 + z_3 = 2 \end{cases}$$

$$\begin{cases} z_1 = 6 \\ z_2 = 16 - 2z_1 = 16 - 12 = 4 \\ z_3 = 2 - 2 \cdot 6 + 2 \cdot 4 = 2 - 12 + 8 = -2 \end{cases}$$

$$z = Ux \Rightarrow \begin{pmatrix} 6 \\ 4 \\ -2 \end{pmatrix} = \begin{pmatrix} 2 & 3 & 4 \\ 0 & -1 & 2 \\ 0 & 0 & -2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

$$\begin{cases} 2x_1 + 3x_2 + 4x_3 = 6 \\ -x_2 + 2x_3 = 4 \\ -2x_3 = -2 \end{cases}$$

$x_3 = 1$

$x_2 = 2x_3 - 4 = 2 - 4 = -2$

$x_1 = (6 - 3(-2) - 4(1))/2 = 4$

$$\Rightarrow \boxed{\begin{matrix} x_1 = 4 \\ x_2 = -2 \\ x_3 = 1 \end{matrix}}$$

$A = LU$

$\det(A) = \det(LU) = \det(L)\det(U)$

$= 1 \cdot (2)(-1)(-2)$

Factorización QR

$A = QR$

Q matriz unitaria

R matriz triangular

$Q = (f_1 \; f_2)$

f_i columna i-ésima de Q

$\langle f_i ; f_j \rangle = 0$

$\langle f_i ; f_i \rangle = 1$

$QQ^T = I$

Ejemplo

$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ $a_1 = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$ $a_2 = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$

$\det(A) = 4 - 6 \neq 0 \therefore$ las columnas son LI

$\therefore \{a_1 ; a_2\}$ forman una base

$u_1 = a_1$

$$\mathrm{Proj}_B(A) = \frac{\langle A ; B\rangle}{\langle B ; B\rangle} B$$

$u_2 = a_2 - \mathrm{Proj}_{u_1}(a_2)$

$u_3 = a_3 - \mathrm{Proj}_{u_1}(a_3) - \mathrm{Proj}_{u_2}(a_3)$

$\langle u_1 ; u_2 \rangle = 0$

$\langle u_1 ; u_3 \rangle = 0$

$\langle u_2 ; u_3 \rangle = 0$

$$\boxed{u_k = a_k - \sum_{j=1}^{k-1} \mathrm{Proj}_{u_j}(a_k)}$$

$\vec{e}_1 = \frac{u_1}{\|u_1\|}$ $\quad \vec{e}_2 = \frac{u_2}{\|u_2\|}$ ...

$u_1 = a_1 = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$

$$u_2 = a_2 - \mathrm{Proj}_{u_1}(a_2) = \begin{pmatrix} 2 \\ 4 \end{pmatrix} - \frac{\langle a_2 ; u_1 \rangle}{\langle u_1 ; u_1 \rangle} u_1 =$$

$$= \begin{pmatrix} 2 \\ 4 \end{pmatrix} - \frac{\langle \begin{pmatrix} 2 \\ 4 \end{pmatrix} ; \begin{pmatrix} 1 \\ 3 \end{pmatrix} \rangle}{\langle \begin{pmatrix} 1 \\ 3 \end{pmatrix} \begin{pmatrix} 1 \\ 3 \end{pmatrix} \rangle} \begin{pmatrix} 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \end{pmatrix} - \frac{14}{10} \begin{pmatrix} 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \end{pmatrix} - \frac{7}{5} \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

$$= \begin{pmatrix} 3/5 \\ -1/5 \end{pmatrix}$$

$u_1 = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$ $\quad u_2 = \begin{pmatrix} 3/5 \\ -1/5 \end{pmatrix}$

<u>CHECK</u>

$$\langle u_1 ; u_2 \rangle = \langle \begin{pmatrix} 1 \\ 3 \end{pmatrix} \begin{pmatrix} 3/5 \\ -1/5 \end{pmatrix} \rangle = \frac{3}{5} - \frac{3}{5} = 0 \quad ✓$$

$\therefore \quad u_1 \perp u_2$

$$e_1 = \frac{u_1}{\|u_1\|} = \frac{u_1}{\sqrt{\langle u_1 ; u_1 \rangle}} = \begin{pmatrix} 1 \\ 3 \end{pmatrix} \cdot \frac{1}{\sqrt{10}}$$

$$e_2 = \frac{u_2}{\|u_2\|} = \frac{u_2}{\sqrt{\langle u_2, u_2\rangle}} = \frac{1}{5}\begin{pmatrix}3\\-1\end{pmatrix} \cdot \frac{1}{\sqrt{\frac{9}{25}+\frac{1}{25}}} = \frac{1}{5}\begin{pmatrix}3\\-1\end{pmatrix}\frac{1}{\sqrt{10/25}}$$

$$= \frac{1}{5}\begin{pmatrix}3\\-1\end{pmatrix}\frac{5}{\sqrt{10}} \Rightarrow \quad e_2 = \frac{1}{\sqrt{10}}\begin{pmatrix}3\\-1\end{pmatrix}$$

$$Q = \begin{pmatrix} e_1 & e_2 \end{pmatrix} = \frac{1}{\sqrt{10}}\begin{pmatrix}1 & 3\\ 3 & -1\end{pmatrix}$$

$$A = QR \Rightarrow \quad Q^T A = \underbrace{Q^T Q}_{I} R \Rightarrow \quad R = Q^T A$$

$$R = \frac{1}{\sqrt{10}}\begin{pmatrix}1 & 3\\ 3 & -1\end{pmatrix} \cdot \begin{pmatrix}1 & 2\\ 3 & 4\end{pmatrix} = \frac{1}{\sqrt{10}}\begin{pmatrix}10 & 14\\ 0 & 2\end{pmatrix}$$

$$A = \underbrace{\boxed{\frac{1}{\sqrt{10}}\begin{pmatrix}1 & 3\\ 3 & -1\end{pmatrix}}}_{Q}\ \underbrace{\boxed{\frac{1}{\sqrt{10}}\begin{pmatrix}10 & 14\\ 0 & 2\end{pmatrix}}}_{R}$$

$$\boxed{\begin{array}{c}\text{verificar que}\\ Q^T Q = I\\ Q Q^T = I\end{array}}$$

Aplicación:

$$A x = b$$

$$Q \underbrace{R x}_{z} = b \Rightarrow \quad Q z = b$$

vector de incognitas

$$z = Q^T b = \bar{b}$$

$$z = \bar{b} \Rightarrow \quad \underbrace{R}_{\text{sistema escalonado}} x = \bar{b}$$

$$A = \begin{pmatrix} 1 & 1 \\ 2 & 3 \\ 2 & 1 \end{pmatrix} \qquad a_1 = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} \quad a_2 = \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix}$$

$$A \in \mathbb{R}^{3\times 2} \qquad Q \in \mathbb{R}^{3\times 2} \qquad R \in \mathbb{R}^{2\times 2}$$

$$u_1 = a_1 = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$$

$$u_2 = a_2 - \mathrm{Proj}_{u_1}(a_2) = \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix} - \frac{\langle u_1\, a_2 \rangle}{\langle u_1\, u_1 \rangle} u_1 = \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix} - \frac{\left\langle \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix} \right\rangle}{\left\langle \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} \right\rangle} \Big(\tfrac{1}{2}\ \ldots$$

$$= \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix} - \frac{1+6+2}{1+4+4} \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix} - \frac{9}{9} \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$$

$$\boxed{u_1 = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} \quad u_2 = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}}$$

Verificar que son ortogonales

$$e_1 = \frac{u_1}{\| u_1 \|} = \frac{u_1}{\sqrt{\langle u_1 ; u_1 \rangle}} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} \frac{1}{\sqrt{9}} = \frac{1}{3} \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$$

$$e_2 = \frac{u_2}{\| u_2 \|} = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} \frac{1}{\sqrt{2}}$$

$$\Rightarrow \boxed{Q = \begin{bmatrix} 1/3 & 0 \\ 2/3 & 1/\sqrt{2} \\ 2/3 & -1/\sqrt{2} \end{bmatrix}}$$

$$R = Q^T A = \begin{bmatrix} 1/3 & 2/3 & 2/3 \\ 0 & 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 2 & 3 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 9/3 & 9/3 \\ 0 & \sqrt{2} \end{bmatrix} \boxed{= \begin{bmatrix} 3 & 3 \\ 0 & \sqrt{2} \end{bmatrix}}$$

$$Q^T Q = \begin{bmatrix} 1/3 & 2/3 & 2/3 \\ 0 & 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 1/3 & 0 \\ 2/3 & 1/\sqrt{2} \\ 2/3 & -1/\sqrt{2} \end{bmatrix} = \begin{bmatrix} 9/9 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$QQ^T = \begin{bmatrix} 1/3 & 0 \\ 2/3 & 1/\sqrt{2} \\ 2/3 & -1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 1/3 & 2/3 & 2/3 \\ 0 & 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix}$$

Ejemplo

$$A = \begin{bmatrix} 1 & -2 & 1 \\ -1 & 3 & 2 \\ 1 & -1 & -4 \end{bmatrix} \qquad a_1 = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \quad a_2 = \begin{pmatrix} -2 \\ 3 \\ -1 \end{pmatrix} \quad a_3 = \begin{pmatrix} 1 \\ 2 \\ -4 \end{pmatrix}$$

$$u_1 = a_1 = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$$

$$u_2 = a_2 - \text{proj}_{u_1}(a_2) = a_2 - \frac{\langle a_2\, u_1 \rangle}{\langle u_1\, u_1 \rangle} u_1 = \begin{pmatrix} -2 \\ 3 \\ -1 \end{pmatrix} - \frac{\left\langle \begin{pmatrix} -2 \\ 3 \\ -1 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ -4 \end{pmatrix} \right\rangle}{\left\langle \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \right\rangle} \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$$

$$= \begin{pmatrix} -2 \\ 3 \\ -1 \end{pmatrix} - \frac{-6}{3} \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

$$u_3 = a_3 - \text{Proj}_{u_1}(a_3) - \text{Proj}_{u_2}(a_3)$$

$$= \begin{pmatrix} 1 \\ 2 \\ -4 \end{pmatrix} - \frac{\left\langle \begin{pmatrix} 1 \\ 2 \\ -4 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \right\rangle}{\left\langle \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \right\rangle} \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} - \frac{\left\langle \begin{pmatrix} 1 \\ 2 \\ -4 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\rangle}{\left\langle \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right\rangle} \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 \\ 2 \\ -4 \end{pmatrix} - \frac{(-5)}{3} \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} - \frac{(-2)}{2} \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = \frac{4}{3} \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}$$

(Tarea) verificar

$\langle u_1 ; u_2 \rangle$; $\langle u_1 ; u_3 \rangle$; $\langle u_2 ; u_3 \rangle$

$$e_1 = \frac{u_1}{\|u_1\|} = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \qquad e_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \qquad e_3 = \begin{pmatrix} \sqrt{2}/\sqrt{3} \\ 1/\sqrt{6} \\ -1/\sqrt{6} \end{pmatrix}$$

$$Q = \begin{bmatrix} 1/\sqrt{3} & 0 & \sqrt{2}/\sqrt{3} \\ -1/\sqrt{3} & 1/\sqrt{2} & 1/\sqrt{6} \\ 1/\sqrt{3} & 1/\sqrt{2} & -1/\sqrt{6} \end{bmatrix}$$

$$R = Q^T A = \cdot \; \cdot \; \cdot \; \cdot \; \cdot$$

Verificar $\quad Q^T Q = I$

$$Q Q^T = I$$